How to Use the 3D Tree Map

- Click the crosshairs to enable your location and use the plus/minus icon (bottom-right corner of your screen) or pinch your fingers to zoom in and out.
- Use the home icon (top-right corner of your screen) to return to the default view.
- The compass tool (top-left of your screen) orients the map view; press this at any time to return to the default north-facing view.
- Navigate through the grounds and click on the tree icons to learn more about each tree. Be sure to scroll down through the pop-up window to learn about each tree's features.
    - ***Dooley Trees:*** Trees marked as Dooley Trees will be the more opaque icons. These trees were planted during the time the Dooley's lived here, so they are 100+ years old! Pop-up information for Dooley Trees includes ecosystem services and structural features.
    - ***All Trees:*** These trees are all the non-Dooley trees, which consist of trees planted after the Dooley's left Maymont to the City of Richmond. All trees will appear as the more transparent icons. Pop-up information for these trees is limited to botanical catalog information at this time.

How to Use the 2D Tree Map

- Click the crosshairs to enable your location and use the plus/minus icon (bottom-right corner of your screen) or pinch your fingers to zoom in and out.
- Navigate through the grounds and click on the tree icons to learn more about each tree. Be sure to scroll down through the pop-up window to learn about each tree's features.
    - ***Dooley Trees:*** Trees marked as Dooley Trees will have a white tree symbol. These trees were planted during the time the Dooley's lived here, so they are 100+ years old! Pop-up information for Dooley Trees includes ecosystem services and structural features.
    - ***Champion Trees:*** Trees marked with a ★ are *National Champions*, and those marked with a ☆ are *State Champions*. Trees are measured using circumference, height, and average crown spread. Champion titles are always changing as new trees are discovered and old trees die or lose limbs. Trees that have lost their Champion title are referred to as Legacy Trees. Check out the National Register of Champion Trees and the Virginia Big Tree Register for current status updates.
    - ***All Trees:*** These trees are all the non-Dooley trees, which consist of trees planted after the Dooley's left Maymont to the City of Richmond. They are represented by white dots. Pop-up information for these trees is limited to botanical catalog information at this time.

Terminology and Methods

Ecosystem Services

- *Carbon storage:* Amount of carbon stored in the woody vegetation.
    - Predicted from the biomass of each tree. Open-grown trees are multiplied by 0.8 to account for lower biomass experienced in maintained areas.
- *Carbon Sequestration:* Removal of carbon dioxide (CO2) from the air.
    - Calculated by adding the average diameter growth of the taxonomic group to the existing tree diameter to extract annual sequestration values.
- *Avoided Runoff:* Runoff is the flow of water over land and into nearby water bodies
    - Estimated using local values of expected annual surface runoff.
- *Pollution Removal:*
    - Air pollution removal is calculated from hourly tree-canopy resistances for ozone, sulfur, and nitrogen dioxides. For carbon monoxide and particulate matter, removal rates were based on average values and adjusted based on leaf structure.

Structure and Composition

- *Leaf Area:* The total area of all leaves on a tree.
- *Canopy Cover:* Ratio of the ground cover to ground cover with the potential for shade.
- *Leaf Biomass:* Total dry or wet weight of all leaves on a tree.
- *Crown Width:* The length between the furthest extruding branches of a tree.
- *Crown Height:* The height from below the lowest branch to the top of the tree.
- *Replacement Value:* A tree's replacement value is based on the physical resource only. The valuation includes tree species, diameter, condition, and relevant local information.

GIS Botanical Information

- *Tree ID:* A 6-digit code used to identify each tree in the botanical catalog consisting of a 3-digit species code, species type, and tree number
- *Botanical Name:* Scientific Latin name of a plant using its genus and species for identification.
- *Common Name:* Unofficial name assigned to a plant that can vary by region and apply to multiple different species.
- *Family:* A taxonomic rank used to group together plant genera that share evolutionary ancestors and similar structural traits.

Attributions

Maymont Staff

Authors:

- Sara Reid Lewis, Sean Proietti

Data Collection:

- Sara Reid Lewis, Shaniya Taylor, Sean Proietti, Aubrey Koob, Janai Williams

Licensing:

Data Analysis & Design Tools:

- Esri Inc. 2024. ArcGIS Pro (version 3.4). Computer software. Redlands, CA: Esri Inc.
- National Champion Tree Program. Champion Tree Registry. University of Tennessee Institute of Agriculture, School of Natural Resources.
- Virginia Big Tree Program. Virginia Big Tree Register. Virginia Tech, Department of Forest Resources & Environmental Conservation.

2024 VCU

Authors:

- Nicholas Hanna, Sean Proietti, Catherine Viverette, and Jennifer Cimielli

Data Collection:

- Emily Sharp, Becca Hoover, Logan Andrews, Paaru Raj

Funding:

- Student Research funded by VCU's Rice Rivers Center

Licensing:

Data Analysis Tools:

- i-Tree Streets User's Manual v6.0.35 (n.d.). Retrieved September 25, 2024, from https://www.itreetools.org/tools/i-tree-eco
- "R Core Team (2024). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria."

iTree Written Report:

- Click to see full report